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A METHOD FOR IONIZATION OF A FLUID

Abstract

In a method for ionization of a fluid the fluid is conveyed inside of an elongated container in a gaseous state, and a first pair of electrodes are arranged in the container opposite each other and at a distance from each other in a transverse direction of the elongated container. A voltage is supplied to each electrode in the first pair of electrodes so that electric discharges take place from the electrodes for ionization of the fluid, the fluid is supplied with a flow rate to an inlet of the container for deflecting at least a part of the electric discharges downstream in a fluid flow direction and providing a magnetic field in the vicinity of the first pair of electrodes for affecting a structure of the discharges for supporting the ionization of the fluid.

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Background/Summary

BACKGROUND AND SUMMARY

[0001] The invention relates to a method for ionization of a fluid in a gaseous state.

[0002] Ionization is a process by which an atom or a molecule acquires a negative or positive charge by gaining or losing electrons, often in conjunction with other chemical changes. The resulting electrically charged atom or molecule is called an ion.

[0003] The technical field of the invention regards ionization of the fluid by subjecting the fluid to an electric arc.

[0004] The fluid in gaseous state used as input to the ionization method may be air. When the electric arc is sufficiently strong, conditions are created for the gas to become separated into positive ions and electrons, wherein the air is ionized.

[0005] One area of application of an ionized gas is to clean a fluid, such as a gas, that may be air, or a liquid such as industrial process liquids and all kinds of water and wastewater and other liquids. The ionized gas may be used for eliminating organic and mineral impurities or pollutants. Such organic matter may be bacteria, viruses, other harmful microorganisms, and some organic chemical substances. Also, for separation by sedimentation of the inorganic or mineral substances such as metal.

[0006] The disclosure in this chapter should not be regarded as any admittance of prior art.

[0007] WO2018/211309 discloses an electric arc ionization reactor and a method for generating ozone by using air. The reactor is elongated and has an inner circular cross section shape. An inlet for entry of air is provided at a first end of the elongated reactor and an outlet is provided at a second end of the elongated reactor. A pair of needle electrodes are arranged opposite each other and at a distance from each other in a transverse direction of the elongated reactor. An alternating current with high voltage is supplied to the electrodes for generating an arc between the electrodes.

[0008] Buntat, Zokafle; Ozone generation using electrical discharges; A comparative study between Pulsed Streamer Discharge and Atmospheric Pressure Glow Discharge (2005). This thesis deals with an investigation into atmospheric pressure glow discharge and pulsed streamer discharge techniques of ozone generation, in an attempt to compare their performances in the generation of a high concentration and high yield of ozone. The art includes the comparison of the different methods of utilization of the corona discharge in atmospheric pressure by use of dielectric plates with the distance of maximum 1 mm from each other.

[0009] US20020170817 discloses the generating of a corona or other electric discharge and provides for the passing of a gas through the corona to effect ionizing, creating of ozone or the like. According to various methods of the invention, a corona discharge (or other electric discharge) is created, a gas is passed through the corona discharge; mixing of the gas may be provided by motionless mixing technique for one or more purposes, such as to assure maximum exposure of the gas to the corona discharge, to provide uniform temperature of the gas, to cool the corona generator, etc.

[0010] JP0761801 discloses an ozonizing unit providing a high ozone concentration by connecting a high-frequency power source across prescribed electrodes, carrying out the corona discharge and regulating the electric current while making the produced ozone gas flow in a spiral form and

preventing ozone from being destroyed.

[0011] U.S. Pat. No. 4,960,569A discloses corona discharge ozonator is provided that comprises a first electrode, a second electrode and a dielectric material disposed between the electrodes. An ozonization chamber is formed between one of the electrodes and the dielectric material and defines a fluid flow path. A plurality of thermally-conducting solids are within the fluid flow path.

[0012] U.S. Pat. No. 6,451,208 discloses a device for applying electrostatic and magnetic fields to a fluid includes an outer conduit and an inner conduit forming a fluid passageway therebetween. The inner conduit is connected to a DC power source and the outer conduit along with electrode needles in electrical communication therewith are connected to ground. A baffle is positioned within the passageway to impart a spiral motion to the fluid flowing therein.

[0013] It is desirable to provide a method for ionizing a fluid flow, which creates conditions for a high ionization efficiency.

[0014] According to an aspect of the invention, a method for ionization of a fluid is provided, wherein the fluid is conveyed inside of an elongated container in a gaseous state, wherein an inner surface of a wall of the elongated fluid container has a circular shape in a cross section transverse to a longitudinal direction of the container, wherein an inner diameter of the container is in a range of 10-50 mm, wherein a first pair of electrodes are arranged in the container opposite each other and at a distance from each other in a transverse direction of the elongated container, wherein the method comprises the steps of supplying a voltage to each electrode in the first pair of electrodes in a range of 2 to 15 kV, especially in a range of 5 to 10 kV, and preferably around 7,5 kV so that electric discharges take place from the electrodes for ionization of the fluid, supplying the fluid with a flow rate in a range of 5-80 litre/min to an inlet of the container for deflecting at least a part of the electric discharges downstream in a fluid flow direction and providing a magnetic field in the vicinity of the first pair of electrodes for affecting a structure of the discharges for supporting the ionization of the fluid.

[0015] A selection of a certain dimension range of a container and a specific combination of parameters and certain parameter ranges for operation of the device has turned out to provide for a high ionization efficiency in a cost-efficient way. The cost-efficiency may be in production and/or operation of the device. The method creates conditions for creating a configuration of the first arc structure that is especially effective in ionization of the fluid.

[0016] According to one embodiment example, the method comprises the step of high frequency charging the electrodes in the first pair of electrodes to an extent that the electric discharges comprise a plurality of independent semi arc structures formed at the same time from each one of the electrodes.

[0017] More specifically, both electrodes exchange electrons/positrons with the environment of the respective electrode eventually leading to a discharge and creation of independent semi arc structures in the vicinity of the respective electrode, in interaction with the conveyed fluid. The term “semi” here meaning that the arcs extend from the respective electrode a distance into the container, but there are no continuous arcs extending between the electrodes in the first pair.

[0018] The method creates conditions for creating electric discharges forming a configuration of a first independent semi arc structure that is especially effective in ionization of the fluid. It may be achieved by supplying a voltage of a certain magnitude to the first pair of electrodes so that both electrodes simultaneously are of the same charge at each moment and providing the fluid in a fluid flow rate matching the magnitude of the voltage. More specifically, the method creates conditions for creating a configuration of the first independent semi arc structure downstream of the electrodes that covers a cross section of the container to a large extent and more specifically may cover a space in the shape of a hemisphere, which in turn creates conditions for ionization of the fluid passing the first arc structure to a large extent. The first independent semi arc structure created may comprise a specific type of arcs, that may be called “Ario-arcs” or “Ario discharge” (Arc-Rotary-Ionization-Orbitals), having certain characteristics, such as a plurality of arcs, permanency of the

arcs and stability of the arcs.

[0019] In other words, the first arc structure may be configured to cover a substantial part of the container in a cross section so that when a stream of atoms/molecules in the fluid flow is conveyed past the electrodes, it makes it difficult for the atoms to pass without getting ionized.

[0020] According to one example, the discharges from the electrodes form “half-arcs” extending from each one of the electrodes to less than or about half-way into the container with regard to a central axis of the container affecting substances passing through the space between the two electrodes.

[0021] According to a further example, at least one arc at a time in the first independent semi arc structure is permanent and continuous.

[0022] According to one example, the first independent semi arc structure comprises an arc having a zigzag shape in the form of saw teeth due to the electrons push each other away due to the same electric charge. In the tips of the saw shaped arc parts, the ionization energy availability is higher (ionization happens relatively easier there) because the electrons are more excited there.

[0023] The wording “emit” electrons/positrons from the electrodes may alternatively be termed “discharge” electrons/positrons.

[0024] According to one example, the method comprises the step of creating the plurality of independent semi arc structures so that it comprises a first set of semi-arcs that are deflected downstream from the first pair of electrodes by the fluid flow and a second set of semi-arcs extending upstream from the first pair of electrodes by the effect of the magnetic field.

[0025] In other words, magnetic bridges (paths made by the magnets for the electrons) may be created in the container in the vicinity of the first pair of electrodes. The magnetic field has an effect on the electrons in the fluid flow passing by so that the electrons are maintained for a few more split seconds in the ionization area. This provides better paths for the creation of a greater number of arcs the first arc structure. Further, by applying a magnetic field a similar yield may be achieved with lower power consumption.

[0026] According to a further embodiment example, the method comprises the step of charging each one of the electrodes in the first pair of electrodes so that they are simultaneously negatively or positively charged creating such a potential difference between each one of the electrodes and an environment of the respective electrode that electric discharges takes place from each one of the electrodes, wherein the method comprises the step of conveying the fluid in a gaseous state inside the container past the first pair of electrodes in the environment of the respective electrode during said charging for ionization of the fluid.

[0027] The method may be used for production of ROS (reactive oxygen species) and some other substances. The fluid used may here be air. After ionization, the fluid comprises a mixture of ROS (reactive oxygen species), such as Oxygen (O_2), Superoxide anion (O_2^-), Peroxide (O_2^{2-}), Hydrogen Peroxide (H_2O_2), Hydroxyl radical (OH) and Hydroxyl ion (OH^-). This is a homogenous mixture of the ROS, and the mixture is substantially stable and radicals that have a relatively higher half-life. It may be maintained stable to be used in a downstream application, such as a tank for cleaning of an industrial process fluid. According to one example, the process fluid is a cutting fluid resulting from an industrial cutting operation.

[0028] According to one example, the method comprises the step of supplying a magnitude of the voltage to the first pair of electrodes that selective ionization is achieved. For example, Oxygen ionizes at a lower energy than Nitrogen. More specifically, an ionization energy of about 1400 KJ/mol would ionize Oxygen and not Nitrogen. The ionization energy is carefully controlled for a specific application for preferably ionizing all the elements to Oxygen and not Nitrogen and more atomic numbers, to avoid producing NO_x (NO_3 - HNO_3) and subsequently odour. The voltage range supplied to the electrodes is chosen in a way that the available energy for the ionization of the gaseous elements is high enough to ionize oxygen but not high enough to ionize the Nitrogen.

[0029] Even in a long-term use of the device and depreciation of the power source (such as

transformer), no undesired by-products are produced, because the power would still end up below the first ionization energy of Nitrogen. This is one benefit of this art and is what distinguishes it from many prior arts ranging from UV and Ozone generators, which ionize Nitrogen (first ionization) and Oxygen (second ionization) and therefore produce the mentioned undesired by-products. For example, as the UV lamp depreciates, the wavelength of the emitted electromagnetic wave alters and that alteration, affect the availability of the ionization energy.

[0030] According to a further embodiment example, the method comprises the step of supplying such a magnitude of the voltage to the electrodes that the first semi arc structure comprises a plurality of arcs. According to one example, there are several streams of electrons flowing from each one of the electrodes in different paths, which form different arcs.

[0031] According to one example, the magnitude of the voltage and the fluid flow rate are matched so that at least one semi-arc in the first arc structure is permanent/continuous. The flow direction also affects bending of the semi arcs in a downstream direction.

[0032] According to a further embodiment example, the inner wall of the elongated fluid container has a diameter in a range of 10-30 mm and preferably in the range of 15-25 mm.

[0033] According to a further embodiment example, the electrodes in the first pair of electrodes are arranged at a distance from each other in a range of 2-15 mm, especially in a range of 2-10mm and preferably in a range of 2-4 mm.

[0034] According to one example, for a container with a diameter in the range 15-25 mm, the distance between the electrodes is in the range of 2-4 mm. In this way, corrosion is minimized and the lifespan of the device is thereby increased and/or it may be more cost-effective in use in low maintenance by substituting the electrodes with the new ones less frequently. According to an alternative or complement, the electrodes are provided with chemical coatings with materials such as nano-Titanium dioxide or nano-platinum or any other material that increases the corrosion resistance of the electrodes.

[0035] According to a further embodiment example, each one of the electrodes in the first pair has an elongated shape with a pointy end and wherein the electrodes are arranged so that the pointy ends face each other. According to one example, the electrodes in the first pair of electrodes are straight and of a rod-type with a pointy end (like a needle) and arranged in-line with each other.

[0036] According to a further embodiment example, each one of the electrodes in the first pair has an elongated shape with a pointy end defining an angle in a range of 20-35°. Such sharp tips of the electrodes creates conditions for creating several paths of electrons emitted from the angled surface of the electrode at longitudinally spaced locations. According to one example, the electrodes in the first pair of electrodes are identical.

[0037] According to a further embodiment example, the method comprises the step of supplying such a voltage to the first pair of electrodes that both electrodes have either positive or negative charge simultaneously.

[0038] Consequently, at a certain point in time, both electrodes in the first pair of electrodes are positively charged and therefore emit electrons, wherein the fluid flow in the environment of the respective electrode may be regarded to form a negatively charged region for interaction with the emitted electrons from the electrodes so that a first arc structure is created projecting from each one of the electrodes for ionization of the fluid.

[0039] It has turned out that a maximum energy level of about 1500 kJ/mol may be achieved with 7.5 kV per electrode for enabling the selective ionization as defined above.

[0040] According to one example, a transformer is connected with one of its output terminals to a first electrode in the first pair (and possibly with another one of its terminals to a first electrode in a second pair) for supplying the voltage. According to one example, AC power from a power source, such as the grid, goes to the transformer. Then the transformer turns the voltage from an input of 12 to 220 volts) with a frequency of 50 to 60 Hz to 2*7.5 kV for each pole (associated to one of the electrodes) with a frequency of about 20 kHz by changing the electric charge of the electrodes (AC

current).

[0041] According to one example, supplying power to the electrodes with high frequency and high voltage creates conditions for causing a permanent semi arc structure which is strong enough not to be adversely affected by the passing fluid flow (up to 80 litres per minute). The higher frequency (of transformer) from 10, the greater number of arcs are shaped and seen.

[0042] According to a further embodiment example, the method comprises the step of supplying the fluid flow to the container with a fluid flow rate in a range of 5-40 litre/min and preferably in a range of 8-20 litre/min. It creates conditions for an arc structure that has the following features: a high number of semi arcs, high geographical/spatial coverage of semi arcs in the vicinity of each electrode environment per cross section of the container and an increase in ionization probability (exposure of fluid to semi arc) and ionization efficiency as a result.

[0043] According to a further embodiment example, the method comprises the step of supplying the fluid flow to the inlet of the container in a pulsed manner.

[0044] In an application of cleaning a liquid, such as an industrial process liquid in a downstream tank, the pulsing continuing through the container and in the output flow, prevents overflow of the liquid in some applications in the tank.

[0045] Further, some liquids (for example hardening liquids) are sensitive to aeration processes (such as AOP), with continuous exposure to diffusion of air or a gaseous fluid, they might produce foam, which is unwanted. There is a risk that ingredients of the liquid product being altered leading to that the liquid may lose some of its beneficial features.

[0046] In order to be on the same page in definitions, Pause/pulse ratio is the pause time divided by pulse time which gives a sense of effective exposure (artificial dosage boost for the oxidant created by means of the ionization), where Pulse time is the duration at which the air is being pumped into the input of container/reactor, and subsequently from the output of the reactor to the tank downstream (for liquid applications areas in industry). Pause time is the duration at which the pump has stopped pumping the air to the container/reactor (a lag for releasing the fluid to the tank downstream). Adding this lag would be beneficial for the increase of contact/residence time in the container/reactor but there is a limit to it.

[0047] Acquiring low frequency pulsing may be achieved by means of pneumatic components or special valves.

[0048] According to a further embodiment example, the method comprises the step of supplying the fluid flow in a pulsed manner to the inlet of the container via a pulsing duration in a range of 0,25-3,0 seconds with a pause in between consecutive pulses of 0,25-10,0 seconds, especially via a pulsing duration in a range of 0,4-1,0 seconds with a pause in between consecutive pulses of 0,5-5,0 seconds and more particularly via a pulsing duration at about 0,5 seconds with a pause in between consecutive pulses of about 1,5 seconds.

[0049] One effect of the pulsing is that there will be pressure fluctuations inside the container that leads to a “hammer-drill-effect”, wherein a distance between the molecules is decreased (and thereby a higher likelihood of ionization). Further, the pulsing affects the arcs to greater thickness compared to as if there were no pulsing. Thus, the pulsing creates conditions for a higher ionization efficiency.

[0050] A further effect of the pulsing is that for generating the same yield of oxidants, a lower volume of air is required as feed to the container, which provides for cost efficiency.

[0051] According to one example, a lag of 1,5 second of pausing and 0,5 seconds of pulsing, gives a sweet spot because during the pause time any foam will have the time to reside (liquid gets enough time to rest and reduce the original level in the main tank while at the same time the ionization contact time and thereby ionization efficiency is increased).

[0052] The exemplary sweet spot is defined above as 0.5 seconds of pulse and 1.5 second of pause giving a pause/pulse ratio of $1.5/0.5 = 3$ which is just below 3.33 (0.33 less for safety) providing the highest possible efficiency of ionization while avoiding reaching to 1500 KJ/mol limit (with 2×7.5

kV of 20 kHz of transformation) of available ionization energy. This is thanks to increased contact time during the pause by controlling the flow of the fluid with pulsing.

[0053] According to a further embodiment example, a magnetic field generating arrangement is arranged outside of the container and adapted to provide the magnetic field in the vicinity of the first pair of electrodes for affecting the first arc structure.

[0054] According to a further embodiment example, the method comprises the step of conveying at least a first portion of the fluid along a helical path inside of the container.

[0055] Such a fluid flow pattern allows the fluid to spend more time in the container, which creates conditions for increasing the combination rate and the likelihood of collisions and therefore ionization rate which leads to a higher ionization efficiency. Further, such a flow pattern may cause the fluid flow to arrive at the first arc structure with a direction angled in relation to a longitudinal direction of the container which in turn may cause more molecules to be ionized by the first arc structure. Further, such a flow pattern may cause turbulence in the fluid flow, which in turn may cause more molecules to be ionized by the first arc structure.

[0056] According to a further embodiment example, the method comprises the step of conveying at least a second portion of the fluid along a substantially straight path inside of the container towards a position between the first pair of electrodes. The second portion of the fluid may in this way contribute substantially to push the first arc structure downstream and thereby create conditions for a high coverage of the cross section of the container and thereby a high ionization efficiency.

[0057] The term “semi arc” structure may be regarded as an arc structure between the prior art arc structures of Glow corona and streamer corona. However, since the semi arc structure introduced by this art does not have a ground pole and/or dielectric, it does not fall within the category of corona discharge and therefore it has a specific shape by a specific arrangement of magnetic field (but according to the vision error it may look like a complete arc extending between the electrodes).

[0058] According to a further embodiment example, the method comprises the step of providing a pressure in the container above 1,1 bars during the supply of voltage to the electrodes. According to a preferred example, the method comprises the step of providing a pressure in the container above 1,5 bars during the supply of voltage to the electrodes. According to one example, the method is operated with a pressure in the container in a range of 1,5-2,0 bar. A pressure level above 1,1 bars increases the likelihood of more collisions of matter and therefore a higher ionization efficiency. The pressure required in the container is further dependent on a downstream application, wherein the pressure level may be up to 10 bars.

[0059] Accordingly, the pressure in the container is greater than the atmospheric pressure during the charging of the electrodes, wherein an increased constant pressure in a given volume in the vicinity of each electrode leads to higher chance of collision (of molecules, ions, atoms, electrons and positrons) and ionization around the respective electrode.

[0060] According to a further embodiment example, the method comprises the step of radiating the fluid in the container via a light source. The light-matter interaction provides for the phenomenon of Pair Production. In the area in which the arcs are formed, the interaction between the photons from the light source with the substances passing by the area, leads to emission of waves with different ranges of wavelength depending on the light source. The generated waves enhance the ionization efficiency. Additionally, electrons and positrons are released, and they contribute to the ionization reactions. And the yield per power consumption is increased.

[0061] The light source is preferably arranged outside of the container. It creates conditions for a long life of the light source since it will not be subjected to the interior environment (friction and heat) of the container. The radiation by the light source may radiate the fluid flow provided the container wall is transparent, such as made of glass.

[0062] The light source may be light-emitting diodes (LED) adapted for radiating an ultraviolet (UV) light. According to an alternative, a Xenon lamp may be used. According to one example, the

light sources may be adapted to provide a light intensity in a range of 100-5600 Lumen. The light intensity may be matched to the magnitude of the voltage supplied to the electrodes, wherein a lower voltage may be compensated by a higher light intensity for a certain ionization effect.

[0063] The light source may contribute to a significantly increased ionization efficiency. Tests have shown results of an increased ionization efficiency of up to 40%.

[0064] According to a further example, the ionization device comprises a second pair of electrodes arranged in the container at a distance from the first pair of electrodes. According to one example the distance between the adjacent electrode pairs is at least 30 mm.

[0065] According to a further embodiment example, a second pair of electrodes are arranged in the container opposite each other and at a distance from each other, wherein the second pair of electrodes are arranged at a distance from the first pair of electrodes downstream of the first pair in a fluid flow direction in the container, wherein the method comprises the step of charging each one of the electrodes in the second pair of electrodes so that they are simultaneously positively charged or negatively charged and synchronizing the charging of the first pair of electrodes in relation to the second pair of electrodes so that the second pair of electrodes are negatively charged when the first pair of electrodes are positively charged and vice versa.

[0066] More specifically, the second pair of electrodes are arranged at such a distance from the first pair of electrodes sufficient to avoid interference of arc structures of adjacent electrode pairs in the frequency synchronization. Further, the distance is preferably sufficient for to avoiding any direct complete arc between two electrodes of opposite charge to avoid any increase in amper load.

[0067] According to a further development of the last-mentioned embodiment example, the method comprises the step of connecting a first electrode in the first pair of electrodes and a first electrode in the second pair of electrodes to opposite terminals of a first power supply and connecting a second electrode in the first pair of electrodes and a second electrode in the second pair of electrodes to opposite terminals of a second power supply.

[0068] Further advantages and advantageous features of the invention are disclosed in the following description and in the dependent claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0069] With reference to the appended drawings, below follows a more detailed description of embodiments of the invention cited as examples. In the drawings:

[0070] FIG. 1 is a schematic view of a device for ionization of a fluid according to a first embodiment, wherein a container is shown in a cross section,

[0071] FIG. 2 is a perspective view of a container in FIG. 1,

[0072] FIG. 3 is an enlarged view of a first pair of electrodes in FIG. 1,

[0073] FIG. 4 is a schematic top view of a first arc structure created in the container according to FIGS. 1 and 2 by the first pair of electrodes,

[0074] FIG. 5 is a schematic front view of the first arc structure created in the container according to FIGS. 1 and 2 by the first pair of electrodes,

[0075] FIG. 6 is a schematic view of a device for ionization of a fluid according to a second embodiment, wherein a container is shown in a cross section,

[0076] FIG. 7 is a perspective view of a fluid flow directing unit provided in the container in FIG. 6,

[0077] FIG. 8 is a front view of the fluid flow directing unit in FIG. 7,

[0078] FIG. 9 is a schematic view of a device for ionization of a fluid according to a third embodiment, wherein a container is shown in a cross section,

[0079] FIG. 10 is a cross section view of the container as in FIG. 9 indicating the fluid flow,

[0080] FIG. **11** is a perspective view of the device for ionization of a fluid according to the first embodiment,

[0081] FIG. **12** is a perspective view of a section of a magnetic field generating arrangement provided around the container in FIG. **11**,

[0082] FIG. **13** is a schematic and partly cut front view of the magnetic field generating arrangement in FIG. **11**,

[0083] FIG. **14** is a schematic view in cross section of the container in FIG. **11** showing parts of the magnetic field generated by the magnetic field generating arrangement,

[0084] FIG. **15** is a perspective view of a container according to an alternative design relative to the container in FIG. **2**,

[0085] FIG. **16** is a schematic view of the container as in FIG. **17** in cross section applied in an ionization device according to a fourth embodiment indicating the fluid flow in operation,

[0086] FIG. **17** is a perspective view of a device for ionization of a fluid according to a fifth embodiment,

[0087] FIG. **18** is a graph indicating an example of a pulsed flow supplied to the ionization device,

[0088] FIG. **19** is a graph indicating available energy for ionization for different pause/pulse ratios,

[0089] FIG. **20** is a partly cut and exploded perspective view of an arrangement for ionization of a fluid, and

[0090] FIG. **21** is a schematic view of a device for ionization of a fluid according to an alternative to the first embodiment.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0091] FIG. **1** is a schematic view of a device **2** for ionization of a fluid in a gaseous state according to a first embodiment. The fluid in gaseous state will in the following be referred to as a gas. According to one example, the gas is air. The ionization device **2** comprises a container **4**. The container **4** is illustrated in a cross section in a horizontal plane through its centre axis. The container **4** has an elongated shape. The container **4** has a rounded cross section shape and more specifically a circular cross section shape. Further, the cross section of the container **4** is constant along a significant portion of the length of the container. Further, ends **6**, **8** of the container **4** in the longitudinal direction have a rounded and more specifically half-spherical shape. A wall **10** of the container **4** defines an inner chamber **12**. An inner surface of the wall **10** of the elongated container **4** has a diameter of about 20 mm. FIG. **2** is a perspective view from above of the container **4** in FIG. **1**.

[0092] The container wall **10** is formed in glass. The container may be formed in two identical container parts with a delimitation in a plane through a center axis of the container **4**. According to an alternative, the container **4** is formed in one-piece with a cap at one end.

[0093] Further, an inlet **14** is provided at a first end **6** of the container **4** in its longitudinal direction and an outlet **16** is provided at a second end **8** of the container **4** in its longitudinal direction for conveying a gas flow from the inlet **14** to the outlet **16**. Each one of the inlet **14** and

[0094] the outlet **16** has a generally tube shape. An axis of the inlet **14** has a main direction in parallel with the longitudinal direction of the container **4** and is arranged in-line with a longitudinal centre axis **17** of the container. Likewise, an axis of the outlet **16** has a main direction in parallel with the longitudinal direction of the container **4** and is arranged in-line with the longitudinal centre axis **17** of the container. The container **4** has a length in a range of 100-120 mm excluding the inlet **14** and outlet **16**.

[0095] The ionization device **2** further comprises a first pair **18** of electrodes **20**, **22** arranged in the container **4** opposite each other and at a distance from each other. The electrodes **20**, **22** are arranged perpendicularly relative to the longitudinal direction of the container **4**. Further, the container **4** is arranged in a way that its longitudinal direction is in a horizontal plane. More specifically, the electrodes **20**, **22** are arranged so that they extend in a horizontal plane. The electrodes **20**, **22** are shown in an enlarged view in FIG. **3**. The electrodes **20**, **22** are arranged at a

distance y from each other in a range of 2-4 mm. Further, each one of the electrodes **20**, **22** in the first pair **18** has an elongated shape with a circular cross section and a pointy end **24**, **26**. The electrodes are arranged so that the pointy ends **24**, **26** face each other. More specifically, each one of the electrodes **20**, **22** in the first pair **18** has an elongated shape with a pointy end **24**, **26** defining an angle α in a range of 20-35°. In other words, each one of the electrodes **20**, **22** has a sharp tip. More specifically, the electrodes **20**, **22** in the first pair **18** are straight and arranged in-line with each other. More specifically, the electrodes **20**, **22** in the first pair **18** are in the form of rods. The electrodes **20**, **22** in the first pair **18** may be termed needle electrodes. The electrodes **20**, **22** in the first pair **18** are formed in a metallic material and more precisely in the material tungsten (also called wolfram) as an example.

[0096] According to physics law, when charging an element, the charged parts accumulate in any sharp edges of the element. Accordingly, charged parts will be highly accumulated in the sharp edge of the electrode **20**, **22**. In other words, charged parts will have a very high density in the sharp edge, wherein an electric field will be strong in a region of the sharp edge. Further, a highly charged electrode (positive or negative) will have a very high potential in relation to the environment (adjacent the electrode). The potential difference between the electrode and its adjacent environment/surroundings will result in ionization of the matter in the environment in the vicinity of the respective electrode leading to exchange of electrons/positrons in cycles from higher potential area to lower potential area and vice versa and different types of electric discharge from the electrode may take place. This phenomenon may be similar to a Tesla coil.

[0097] Accordingly, the design of the electrodes **20**, **22** with sharp tips especially with an acute angle from 20 to 35 degrees (preferably 22 degrees to create a greater number of semi arcs as well as increasing the life span of the electrodes) **24**, **26** creates good conditions for creating electric discharges in the form of electric semi-arcs from the surface of the tip having an inclination relative to the longitudinal direction of the elongate electrode. More specifically, a first set of arcs may be created extending from the electrode tip in a downstream direction. Further, a second set of arcs may be created extending from the electrode tip in an upstream direction. It will be described in more detail below in association with FIG. 4 and FIG. 5. The ionization device **2** further comprises a power supply **28**, **50** adapted for charging each one of the electrodes **20**, **22** in the first pair **18** of electrodes so that they have the same charge simultaneously. In this way, such a potential difference between each one of the electrodes **20**, **22** and an environment of the respective electrode may be created that electric discharges take place simultaneously from each one of the electrodes. Further, the fluid is conveyed in a gaseous state inside the container past the first pair **18** of electrodes in the environment of the respective electrode **20**, **22** during said charging for ionization of the fluid.

[0098] More specifically, the power supply **28**, **50** comprises two transformers **28**, **50**, which are adapted to provide an alternating current of a certain frequency to the electrodes. Accordingly, the power supply **28**, **50** is adapted to supply such a voltage to the first pair **18** of electrodes that both electrodes **20**, **22** are positively charged at the same time and therefore emit electrons. It is schematically shown in a top view in FIG. 4, wherein the arrows **30**, **31** indicate paths of electrons emitted from the tips of the electrodes **20**, **22**. Further, the container **4** is adapted for conveying the gas in a flow past the first pair **18** of electrodes, wherein the gas flow may be regarded as a negatively charged region **32** between the electrodes **20**, **22** for interaction with the emitted electrons from the electrodes so that a first arc structure **34** may be created. More specifically, a plurality of electric semi-arcs project from each one of the electrodes **20**, **22** for ionization of the gas. FIG. 5 is a schematic front view of the first arc structure **34** created in the container according to FIG. 4. Further, each arc has a zigzag shape in the form of saw teeth.

[0099] More specifically, each transformer **28**, **50** comprises a primary winding and a secondary winding. Each transformer turns the voltage from an input of 12 to 220 volts with a frequency of 50 to 60 Hz to 2*7.5 kV for each pole (associated to one of the electrodes) with a frequency of about 20 kHz by changing the electric charge of the electrodes (AC current). Accordingly, each

transformer **28, 50** comprises a frequency converter **29, 51** where one of the ground wire functions is to reduce noise.

[0100] It should be noted that the zigzag arc shapes shown in the FIG. **4** and FIG. **5** are only schematically shown. Especially, the semi-arcs are magnified and much bigger than the real size in relation to the size of the electrodes **20, 22**. In reality the zigzags are in microscopic scales. Also, their plurality is much higher than the number of the arcs shown in the figures.

[0101] Each one of the transformers **28, 50** is adapted for supplying an output voltage at a magnitude of around 7,5 kV via each one of its output terminals. Further, each one of the transformers **28, 50** is adapted for supplying the output voltage in a frequency about 20 kHz, wherein the polarity of the electrodes connected to the two output terminals/poles of one transformer will change very fast (every 0.00005 second).

[0102] The ionization device **2** further comprises a magnetic field generating arrangement **304**. It will be described in more detail below in association with FIG. **11-14**.

[0103] The magnetic field generating arrangement **304** comprises a first section **305** arranged upstream of the first pair **18** of electrodes in the longitudinal direction of the container, wherein the first independent semi-arc structure **34** comprises a first set of semi-arcs **334** that are deflected downstream from the electrodes by the gas flow and a second set of semi-arcs **336** extending upstream from the electrodes **20, 22** by the effect of the magnetic field of the first magnetic field generating section **305**. The first set of arcs **334** and the second set of arcs **336** are shown in FIG. **4**. It may be noted that the second set of arcs **336** comprises fewer arcs than the first set of arcs **334** and that the arcs in the second set of arcs **336** have a smaller extension in the longitudinal direction of the container relative to the first set of arcs **334**.

[0104] More specifically, the magnetic field generated by the first magnetic field generating section **305** creates bridges/pathways for arcs also upstream of the electrode pair **18**, see arrows **330** and **331** indicating the electrons emitted from the electrodes **20, 22**. The second set of arcs **336** comprises a plurality of arcs between the electrodes **20, 22**. Further, the arcs have a zigzag/saw-tooth shape.

[0105] It should be noted that the zigzag arc shapes shown in the FIG. **4** and FIG. **5**, are magnified and much bigger than the real size in relation to the size of the electrodes **20, 22**. In reality the zigzags are in microscopic scales. Also, their plurality is much higher than the number of the arcs shown in the figures.

[0106] More specifically, each one of the electrodes **20, 22** is arranged in an opening **36, 38** through the container wall **10**. More specifically, the container comprises a pipe-shaped portion **40, 42** extending in a transverse direction relative to the longitudinal direction of the container **4**. More specifically, the pipe-shaped portion **40, 42** extends perpendicularly relative to the longitudinal direction of the container **4**. The pipe-shaped portions **40, 42** define the openings **36, 38**. More specifically, the pipe-shaped portions **40, 42** are formed in one-piece with the container **4**. More specifically, the electrodes **20, 22** are arranged in the pipe-shaped portions **40, 42** in a gas tight manner for avoiding leakage.

[0107] The ionization device **2** further comprises a second pair **44** of electrodes **46, 48** arranged in the container **4** in a similar way as has been described above with regard to the first pair **18** of electrodes **20, 22**. The second pair **44** of electrodes **46, 48** are arranged at a distance from the first pair **18** of electrodes **20, 22** in the longitudinal direction of the container **4**. Each one of the first pair **18** of electrodes **20, 22** and the second pair **44** of electrodes **46, 48** are arranged at the portion of the container **4** having a constant cross section with a distance between the adjacent electrode pairs of about 30 mm. The power supply **28,50** is adapted for charging each one of the electrodes **46, 48** in the second pair **44** of electrodes so that they have the same charge at the same time. In this way, such a potential difference between each one of the electrodes **46, 48** and an environment of the respective electrode may be created that electric discharges take place from each one of the electrodes separately. Accordingly, the power supply **28, 50** is adapted to supply such a voltage also

to the second pair **44** of electrodes that both electrodes **46, 48** are positively charged at the same time and therefore emit/exchange electrons/positrons.

[0108] The arrangement is adapted to synchronize the charging of the first pair **20** of electrodes **20, 22** in relation to the second pair **44** of electrodes **46, 48** so that the second pair **44** of electrodes **46, 48** are negatively charged when the first pair **20** of electrodes **20, 22** are positively charged and vice versa.

[0109] The two transformers **28, 50** are of the same natural frequency and identical. By arranging the transformers **28, 50** adjacent each other in relative close proximity, their frequency cycles will become synced eventually in a steady state since they would influence each other during operation, due to Hertz and frequency laws. Accordingly, they can work with a synced frequency permanently. Accordingly, this synchronization happens spontaneously as soon as the transformers are turned on. According to an alternative, means may be provided to actively control the synchronization, such as arranging a one-way diode (a diode that synchronize the direction of the current in the same direction-the sinus or cosinus wave) in the path of each exit terminal.

[0110] Accordingly, each one of the transformers has two output terminals/poles, which are connected to the electrodes **20, 22; 46, 48** for charging the electrodes. When the potential reaches an amount that is sufficient for electric discharge, the above-mentioned phenomenon of electric discharge will take place. More specifically, a first electrode **22** in the first pair **18** of electrodes and a first electrode **46** in the second pair **44** of electrodes are connected to opposite terminals of a first transformer **28**. Further, a second electrode **20** in the first pair **18** of electrodes and a second electrode **48** in the second pair **44** of electrodes are connected to opposite terminals of a second transformer **50**.

[0111] The ionization device **2** further comprises a gas flow pumping means **52** for supplying the gas flow from a tank **54** of compressed air to the inlet **14** of the container **4**. More specifically, the gas flow pumping means **52** is adapted for supplying the gas flow at such a rate to the container **4** that it is conveyed past the first pair **18** of electrodes **20, 22** so that at least parts of the first arc structure are deflected downstream from the electrodes **20, 22** in a direction of the gas flow. More specifically, the gas flow pumping means **52** is adapted for supplying the gas flow to the container with a gas flow rate in a range of 10-12 litre/min.

[0112] It may be noted that the device is not limited to an application of a tank for supply of the gas. It can be a compressor using ambient air or an industrial blower, etc.

[0113] Further, the gas flow pumping means **52** is adapted for supplying the gas flow to the inlet **14** of the container **4** in a pulsed manner. The method comprises the step of supplying the gas flow in a pulsed manner to the inlet **14** of the container **4** via a pulsing duration at about 0,5 seconds with a pause between consecutive pulses of about 1,5 seconds, see graph in FIG. **18**.

[0114] The outlet **16** of the container **4** is in fluid communication with a tank **56** comprising a process liquid, such as industrial water or wastewater that have strong aerobic or anaerobic bacteria. A line connecting the outlet **16** with the tank **56** ends in a lower region of the tank **56** so that the ionized gas may be supplied below a surface of the process liquid in order to separate inorganic or mineral substances such as metal by sedimentation or kill the bacteria.

[0115] According to an alternative, the tank **56** is replaced with another device that relates to air purification. The ionized gas exiting the outlet can be directly sprayed to a room to eliminate virus, bacteria, odour, etc.

[0116] A further effect of the pulsing is that a less amount of unionized air (O₂) per volume of output is sent to the tank **56**. Unionized air may risk support the aerobic bacteria to grow and it will compete with the ionized portion of the air. With pulsing, more ionized air compared to unprocessed air (O₂) is sent in the mixture per volume of output fluid.

[0117] FIG. **6** is a schematic view of a device **102** for ionization of a gas according to a second embodiment. The ionization device **102** according to the second embodiment has many parts in common with the ionization device **2** according to the first embodiment. For ease of presentation,

only the main differences will be explained below.

[0118] The ionization device **102** comprises a nozzle **104** arranged in the inlet **14** of the container **4**. The nozzle **4** is adapted for being rotated around an axis in parallel with an axis of the inlet **14** for conveying the gas along a helical path inside of the container **4**. The nozzle **104** comprises an end portion facing the container inner chamber **12** having radially external surfaces defining a generally circular cross section shape that is matched to a dimension of an inner surface of the inlet **14**. Further, the nozzle **104** comprises peripheral through-going channels adapted for creating a helical flow inside of the container **4**.

[0119] The ionization device **102** further comprises a first fluid flow directing unit **106** arranged in the container **4**. The first fluid flow directing unit **106** is arranged downstream of the first pair **18** of electrodes. More specifically, the first fluid flow directing unit **106** is arranged downstream of the second pair **44** of electrodes.

[0120] The first fluid flow directing unit **106** is adapted to compensate for a pressure drop throughout the length of the container **4** by providing a hindrance to the gas flow. In this way, a second arc structure created by the second pair **44** of electrodes may be as strong and disciplined as the first arc structure created by the first pair **18** of electrodes. More specifically, the pressure in the container **4** is maintained or at least not significantly reduced thanks to the first fluid flow directing unit **106**. A distance between the molecules is decreased and the retention time in the container is increased and consequently the ionization efficiency increases. Further, maintaining the pressure at a relatively high level may be important for the delivery of the fluid to the tank **56**, since the liquid in the tank provides a counterpressure that needs to be overcome.

[0121] FIG. 7 is a perspective view of the first fluid flow directing unit **106** provided in the container **4** in FIG. 6. FIG. 8 is a front view of the first fluid flow directing unit **106** in FIG. 7. The first fluid flow directing unit **106** comprises at least one peripheral fluid flow guide channel **108** having an outlet **110** circumferentially displaced relative to an inlet **112** for turning a first part of an incoming fluid flow. The first fluid flow directing unit **106** further comprises a central fluid flow guide channel **114** with an extension substantially in parallel with the longitudinal direction of the elongated container **4** for guiding a second part of the incoming fluid flow substantially in the longitudinal direction of the elongated container **4**.

[0122] More specifically, the first fluid flow directing unit **106** comprises a plurality of circumferentially spaced peripheral fluid flow guide channels **108**, **118**, **120**. Further, the at least one peripheral fluid flow guide channel **108**, **118**, **120** has a substantially larger dimension than the central fluid flow guide channel **114** for conveying a substantially larger part of the incoming fluid flow.

[0123] Further, the first fluid flow directing unit **106** has a rounded peripheral surface **122** substantially corresponding to a curvature of the rounded inner surface of the container **4**, wherein the first fluid flow directing unit **106** is arranged in the container **4** so that the rounded surfaces are in contact with each other in a fluid tight manner.

[0124] More specifically, the first fluid flow directing unit **106** is rigidly connected, such as via weld seams, to the container **4** in an operational position. The first fluid flow directing unit **106** may be formed in a material with the same or similar expansion coefficient as the container wall **10**. According to one example, the first fluid flow directing unit **106** is formed in glass. It creates conditions for rigidly connecting the first fluid flow directing unit **106** to the container **4** in the operational position via welding.

[0125] More specifically, the first fluid flow directing unit **106** comprises a body **124** defining the at least one peripheral fluid flow guide channel **108**, **118**, **120** and the central fluid flow guide channel **114**. More specifically, the first fluid flow directing unit **106** is formed by a one-piece body **124**.

[0126] The at least one peripheral fluid flow guide channel **108**, **118**, **120** is open in a radial direction of the first fluid flow directing unit **106**. More specifically, at least one peripheral fluid

flow guide channel **108, 118, 120** is closed in the radial direction by the wall **10** of the container **4** in FIG. **6**.

[0127] The first fluid flow directing unit **106** comprises sections **126, 128, 130** circumferentially between adjacent peripheral fluid flow guide channels **108, 118, 120**. A radially outer surface of these sections **126, 128, 130** of the first fluid flow directing unit **106** defines a circular shape of substantially the same dimension as the inner surface of the elongated container **4**. A wall of each one of the sections **126, 128, 130** faces in the longitudinal direction of the container **4** for blocking parts of the fluid flow. A total area of the walls of the sections **126, 128, 130** is substantially the same as a cross section area defined by the peripheral fluid flow guide channels **108, 118, 120**.

[0128] The first fluid flow directing unit **106** is adapted for conveying at least a first portion of the fluid along a helical path inside of the container **4** via the at least one peripheral fluid flow guide channel **108, 118, 120**. Further, the first fluid flow directing unit **106** is adapted for conveying at least a second portion of the fluid along a substantially straight path inside of the container via the central fluid flow guide channel **114**.

[0129] FIG. **9** is a perspective view of a device **202** for ionization of a gas according to a third embodiment. The ionization device **202** according to the third embodiment has many parts in common with the ionization device **102** according to the second embodiment. For ease of presentation, only the main differences will be explained below.

[0130] The ionization device **202** comprises a second fluid flow directing unit **206**. The two fluid flow directing units **106, 206** are arranged spaced from each other in the longitudinal direction of the container **4**. More specifically, the two fluid flow directing units **106, 206** are arranged on opposite sides of the first pair **18** of electrodes **20, 22**. More specifically, the two fluid flow directing units **106, 206** are arranged on opposite sides of the first pair **18** of electrodes **20, 22** and the second pair **44** of electrodes **46, 48**. More specifically, the second fluid flow directing unit **206** has a design similar to the design of the first fluid flow directing unit **106** with the difference that the at least one peripheral fluid flow guide channel is turned circumferentially in an opposite direction. Thus, the two fluid flow directing units **106, 206** are identical in dimension but have a mirrored design for the change of direction of the fluid flow. In other words, a first one of the two fluid flow directing units **106, 206** is adapted to turn the fluid flow in a clockwise direction and the other one is adapted to turn the fluid flow in a counter clockwise direction.

[0131] FIG. **10** is a schematic top view of the ionization device **202** as in FIG. **9** indicating the fluid flow paths. The peripheral fluid flow guide channel **108, 118, 120** of the upstream first fluid flow directing unit **206** are adapted to convey a first part of the fluid flow in a helical path **208** inside the container **4**. Further, the central fluid flow guide channel **114** is adapted to convey a second part of the fluid flow in a substantially straight path **210** inside of the container in parallel with the longitudinal direction of the container **4**.

[0132] FIG. **11** is a perspective view of parts of the ionization device **2**.

[0133] The magnetic field generating arrangement **304** is adapted for generating a magnetic field in the vicinity of the first pair **18** of electrodes for affecting the arc structures for supporting the ionization of the gas. The magnetic field generating arrangement **304** is arranged outside of the container **4**. It creates conditions for a long life of the magnetic field generating arrangement **304** since it will not be subjected to the interior environment (friction and heat) of the container **4**.

[0134] The magnetic field generating arrangement **304** comprises at least one magnetic field generating unit **310**. The magnetic field generating unit **310** is formed by an electromagnet **308**. The electromagnet **308** comprises a coil adapted for the passage of electric current. The electromagnet **308** is arranged so that an axis of the coil extends in a radial direction in relation to the container **4**. According to an alternative, the magnetic field generating unit **310** is formed by a permanent magnet. The magnetic field generating unit is adapted for providing a magnetic strength in a range of 20-180 and especially in a range of 20-40 N.

[0135] More specifically, the first magnetic field generating section **305** comprises a plurality of

circumferentially spaced magnetic field generating units **310** around the container. According to the shown example, the first magnetic field generating section **305** comprises six circumferentially spaced magnetic field generating units **310** around the container. The number of magnetic field generating units **310** may of course be altered depending on the application. Further, each one of the circumferentially spaced magnetic field generating units **310** is formed by an electromagnet. According to an alternative, one or several or all of the circumferentially spaced magnetic field generating units **310** may be formed by a permanent magnet.

[0136] Referring now also to FIG. **12** that is a perspective view of the first magnetic field generating section **305**. The first magnetic field generating section **305** comprises a ring-shaped support **312** extending around the container, wherein the ring-shaped support is adapted to hold the circumferentially spaced magnetic field generating units **310** in their operational positions. Each one of the circumferentially spaced magnetic field generating units **310** is arranged so that its axis extends radially outwards from the ring-shaped support **312**.

[0137] FIG. **13** is a partly cut front view of the ionization device **2** in FIG. **1**. The ring-shaped support **312** is arranged in close proximity to an outer wall surface of the container **4**. More specifically, the ring-shaped support **312** has an inner diameter that is somewhat larger than an outer diameter of the container **4**.

[0138] With reference to FIG. **11**, the magnetic field generating arrangement **304** comprises a second section **307**, which is arranged outside of the container and downstream of the first pair **18** of electrodes in the longitudinal direction of the container **4**, wherein the second magnetic field generating section **307** is adapted for generating a magnetic field in the vicinity of the first pair of electrodes for stabilizing the first arc structure. More specifically, the magnetic field created by the second magnetic field generating section **307** is adapted for disciplining the first set of arcs and give them a more coherent arrangement. In other words, the disciplining of the arcs means that the arcs form a more symmetric pattern with certain spacings etc. Also, the magnetic field created by the second magnetic field generating section **307** effects the first set of arcs to increase the quantity of the arcs as well as an increased thickness of the arcs. The second magnetic field generating section **307** is similar in construction and functionality as the first magnetic field generating section **305**.

[0139] Further, the magnetic field generating arrangement **304** comprises a third section **309**, which is arranged outside of the container and upstream of the second pair **44** of electrodes in the longitudinal direction of the container **4**. The third magnetic field generating section **309** is adapted for generating a magnetic field in the vicinity of the second pair **44** of electrodes similar to how the first magnetic field generating section **305** is adapted for generating the magnetic field in the vicinity of the first pair **18** of electrodes and will therefore not be described in any further detail here.

[0140] FIG. **14** is a schematic top view of the magnetic field generating arrangement **304** schematically showing parts **311** of the magnetic fields generated. More specifically, FIG. **14** shows the magnetic fields generated by two opposite magnetic field generating units **310**. Similar magnetic fields are created by each one of the other two pairs of opposite magnetic field generating units **310**.

[0141] FIG. **15** is a perspective view of a container **404** according to an alternative design relative to the container **4** in FIG. **2**. The container **404** differs in relation to the container **4** in FIG. **2** in that it has a further outlet **416**. The further outlet **416** is arranged inclined in relation to the longitudinal direction of the container **404**. More specifically, the further outlet **416** is arranged with an angle of its axis in relation to the axis of the outlet **16** in a range of 30-60° and preferably about 45°. Further, the further outlet **416** is arranged extending from the half-spherical end **8** of the container **404**. The arrangement of two outlets **4**, **404** creates conditions for dividing the ionized gas flow into two separate gas flows to different destinations. According to one example, one of the outlets **16**, **416** may be in fluid communication with the inlet **14** for recirculation of a part of the ionized fluid flow.

[0142] FIG. **16** shows an ionization device **402** according to a fourth embodiment comprising the container **404** according to FIG. **15**. It indicates the fluid flow paths inside of the container **404**. More specifically, the design and position of the first fluid flow directing unit **106** is designed for conveying a first part of the fluid towards the axial outlet **16** and a second part of the fluid towards the further second outlet **416**.

[0143] The ionization device **402** may as an alternative or complement to the first fluid flow directing unit **106** further comprise means for selective guiding of parts of the fluid flow to the outlets **16**, **416**. According to one example, the fluid flow selective guiding means is adapted to attract a negatively charged part of the flow to the further outlet **416**. It may be formed by a further electrode acting as a cathode. Since electrons has a negative charge and some of the ionized molecules/atoms are positively charged, the cathode may attract the negatively charged part of the flow to the further outlet **416** and it can be used for another purpose (for example returning to the inlet **14** or for other purposes). In this way, the axial main output (the target ionization) in the axial outlet **16** is more purified. According to an alternative or complement, an anode may be used to absorb the positively charged part of the flow depending on the purpose.

[0144] FIG. **17** is a schematic and partly cut perspective view of parts of an ionization device **602** according to a fifth embodiment. The ionization device **602** according to the fifth embodiment has many parts in common with the ionization device **2** according to the first embodiment. For ease of presentation, only the main differences will be explained below.

[0145] The ionization device **602** comprises at least one light source **504**, **506**, which is adapted to subject the gas flow in the container for radiation and thereby support the ionization of the gas. The light-matter interaction provides for the phenomenon of Pair Production.

[0146] The light source **504**, **506** is in the shape of a strip extending in the longitudinal direction of the container **4**. The light source **504**, **506** strip has a main extension along a straight line. More specifically, two light sources **504**, **506** are arranged opposite each other, ie spaced with 180°. More specifically, the two light sources **504**, **506** are arranged so that their longitudinal directions are in parallel with each other. More specifically, the strip shaped light source extends along a substantial part of the container **4** and in the shown example substantially along its complete length. The light source **504**, **506** is arranged outside of the container **4**. It creates conditions for a long life of the light source **504** since it will not be subjected to the interior environment (friction and heat) of the container **4**. The radiation by the light source **504** may radiate the fluid flow thanks to the fact that the container wall is transparent.

[0147] The at least one light source **504**, **506** comprises a plurality of light source units arranged in a spaced relationship in a longitudinal direction of the respective strip. The light source **504**, **506** may be light-emitting diodes (LED) adapted for radiating an ultraviolet (UV) light. According to an alternative, a Xenon lamp may be used. According to one example, the light sources **504**, **506** may be adapted to provide a light intensity in a range of 100-5600 Lumen. The light intensity may be matched to the magnitude of the voltage supplied to the electrodes, wherein a lower voltage may be compensated by a higher light intensity for a certain ionization effect.

[0148] According to an alternative, the light source may be a light bulb instead of a strip. Also other shapes and arrangements of the light source may be applicable.

[0149] The ionization device **602** comprises the magnetic field generating arrangement **304** as in FIG. **1**. The ionization device **602** further comprises a support structure **604** for supporting the container **4**, the light sources **504**, **506** and the magnetic field generating arrangement **304** in predefined positions. More specifically, the support structure **604** comprises two blocks **606**, **608**. The blocks **606**, **608** are adapted to be positioned on top of each other. Each block **606**, **608** comprises a receptacle **610**, **612** in the surfaces adapted to face each other. The receptacle **610**, **612** has an elongated extension defining a half-circle in cross section for receiving the container **4**. Further, each block **606**, **608** is designed with internal chambers/receptacles for receiving the light sources **504**, **506** and the magnetic field generating arrangement **304**. Further, each block **606**, **608**

is provided with through holes **614** in a certain configuration for matching each other in order to receive bolts for securing the blocks **606**, **608** to each other. Further, each block **606**, **608** may be adapted to also receive the transformers **28**, **50**.

[0150] FIG. **19** is a graph indicating available energy for ionization for different pause/pulse ratios. As seen in the graph, to any ratio below 3.3 (and more safely below 3) would be beneficial as a pulsing feature. However, 3 is the optimal value. It provides for a highest possible efficiency of ionization while avoiding reaching to 1500 KJ/mol limit (with 2×7.5 kV of 20 kHz of transformation) of available ionization energy. This is thanks to increased contact time during the pause by controlling the flow of the fluid with pulsing.

[0151] FIG. **20** is a partly cut and exploded perspective view of an arrangement **702** for ionization of a fluid. The arrangement **702** comprises the ionization device **2** in FIG. **1** arranged in a casing **714** that has a generally circular cylindrical outer shape. The arrangement **702** comprises a generally flat rectangular wall **718** and wall **720** that is generally half-circular in cross section, which is connected to the flat rectangular wall **718** in a way defining an internal space between the walls **718**, **720**. The ionization device **2** is arranged in the internal space between the walls **718**, **720**. The transformers **28**, **50** are arranged on either side of the container **4** in its longitudinal direction and connected to the electrodes **20**, **22**, **46**, **48** as described above. Further, the transformers are located within the internal space between the walls **718**, **720**.

[0152] FIG. **21** is a schematic view of a device **802** for ionization of a fluid according to an alternative to the first embodiment. The ionization device **802** differs from the first embodiment in the structure of the transformers **828**, **850**. More specifically, a secondary mid-point of a secondary winding is connected to earth.

[0153] It is to be understood that the present invention is not limited to the embodiments described above and illustrated in the drawings; rather, the skilled person will recognize that many changes and modifications may be made within the scope of the appended claims.

[0154] The invention has been described above for an application of cleaning industrial process liquids. According to an alternative, the invention may be used for cleaning wastewater, such as municipal wastewater. According to an alternative, the invention may be used for cleaning air, such as air in buildings. The ionized gas may be used for eliminating organic and mineral impurities or pollutants. Such organic matter may be bacteria, viruses, other harmful microorganisms, and some organic chemical substances.

Claims

1. A method for ionization of a fluid, wherein the fluid is conveyed inside of an elongated container in a gaseous state, wherein an inner surface of a wall of the elongated fluid container has a circular shape in a cross section transverse to a longitudinal direction of the container, wherein an inner diameter of the container is in a range of 10-50 mm, wherein a first pair of electrodes are arranged in the container opposite each other and at a distance from each other in a transverse direction of the elongated container, wherein the method comprises the steps of supplying a voltage to each electrode in the first pair of electrodes in a range of 2 to 15 kV so that electric discharges take place from the electrodes for ionization of the fluid, supplying the fluid with a flow rate in a range of 5-80 litre/min to an inlet of the container for deflecting at least a part of the electric discharges downstream in a fluid flow direction and providing a magnetic field in the vicinity of the first pair of electrodes for affecting a structure of the discharges for supporting the ionization of the fluid.
2. A method according to claim 1, wherein the method comprises the step of high frequency charging the electrodes in the first pair of electrodes to an extent that the electric discharges comprise a plurality of independent semi arc structures formed at the same time from each one of the electrodes.
3. A method according to claim 2, wherein the method comprises the step of creating the plurality

of independent semi arc structures so that it comprises a first set of semi-arcs that are deflected downstream from the first pair of electrodes by the fluid flow and a second set of semi-arcs extending upstream from the first pair of electrodes by the effect of the magnetic field.

4. A method according to claim 1, wherein the method comprises the step of charging each one of the electrodes in the first pair of electrodes so that they are simultaneously negatively or positively charged creating such a potential difference between each one of the electrodes and an environment of the respective electrode that electric discharges takes place from each one of the electrodes, wherein the method comprises the step of conveying the fluid in a gaseous state inside the container past the first pair of electrodes in the environment of the respective electrode during the charging for ionization of the fluid.

5. A method according to claim 1, wherein the inner wall of the elongated fluid container has a diameter in a range of 10-30 mm and preferably in the range of 15-25 mm.

6. A method according to claim 1, wherein the electrodes in the first pair of electrodes are arranged at a distance from each other in a range of 2-15 mm.

7. A method according to claim 1, wherein each one of the electrodes in the first pair has an elongated shape with a pointy end and wherein the electrodes are arranged so that the pointy ends face each other.

8. A method according to claim 7, wherein each one of the electrodes in the first pair has an elongated shape with a pointy end defining an angle in a range of 20-35°.

9. A method according to claim 1, wherein method comprises the step of supplying the voltage to each electrode in the first pair of electrodes in a frequency range of 10-30 kHz.

10. A method according to claim 1, wherein the method comprises the step of supplying the fluid flow to an inlet of the container with a fluid flow rate in a range of 5-40 litre/min.

11. A method according to claim 1, wherein method comprises the step of supplying the fluid flow to an inlet of the container in a pulsed manner.

12. A method according to claim 11, wherein the method comprises the step of supplying the fluid flow in a pulsed manner to the inlet of the container via a pulsing duration in a range of 0.25-3.0 seconds with a pause in between consecutive pulses of 0.25-10.0 seconds.

13. A method according to claim 1, wherein a magnetic field generating arrangement is arranged outside of the container and adapted to provide the magnetic field in the vicinity of the first pair of electrodes for affecting the first arc structure.

14. A method according to claim 1, wherein the method comprises the step of conveying at least a first portion of the fluid along a helical path inside of the container.

15. A method according to claim 1, wherein the method comprises the step of conveying at least a second portion of the fluid along a substantially straight path inside of the container in a longitudinal direction of the container.

16. A method according to claim 1, wherein the method comprises the step of providing a pressure in the container above 1.1 bars during the supply of voltage to the first pair of electrodes.

17. A method according to claim 1, wherein the method comprises the step of radiating the fluid in the container via a light source.

18. A method according to claim 1, wherein a second pair of electrodes are arranged in the container opposite each other and at a distance from each other, wherein the second pair of electrodes are arranged at a distance from the first pair of electrodes downstream of the first pair in a fluid flow direction in the container, wherein the method comprises the step of charging each one of the electrodes in the second pair of electrodes so that they are simultaneously negatively or positively charged and synchronizing the charging of the first pair of electrodes in relation to the second pair of electrodes so that the second pair of electrodes are negatively charged when the first pair of electrodes are positively charged and vice versa.

19. A method according to claim 18, wherein the method comprises the step of connecting a first electrode in the first pair of electrodes and a first electrode in the second pair of electrodes to

opposite terminals of a first power supply and connecting a second electrode in the first pair of electrodes and a second electrode in the second pair of electrodes to opposite terminals of a second power supply.

20. A method according to claim 19, wherein the first power supply and the second power supply are formed by two identical transformers in terms of high voltage and high frequency.

21. A method according to claim 1, wherein the method comprises the step of supplying a magnitude of the voltage to the first pair of electrodes that selective ionization is achieved.
